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Numerical modelling of welding

L.-E. Lindgren *

Division of Computer Aided Design, Luleå University of Technology, S-97187 Luleå, Sweden

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Abstract

The paper describes the application of the finite element method to predict the thermal, material and mechanical effects of welding; Computational Welding Mechanics (CWM). The most common numerical approach used in CWM is only described shortly. The paper focuses on different modelling aspects. The most important modelling issues are the models for heat input and material behaviour. Finally, some recent applications are reviewed and future developments are discussed.
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Keywords: Welding; Finite elements; Thermo-mechanics; Buckling; Cracking; Optimisation

1. Introduction

The history of finite element simulations of the thermal and mechanical behaviour during welding dates back to the 1970s [1–5]. The need to accommodate the effect of the evolving microstructure in the simulations was early recognised. Computational Welding Mechanics (CWM) is therefore concerned with the thermo-mechanical response as well as the changes in material properties due to the thermal cycles in welding.

This paper is focusing on modelling aspects of CWM and also contains an update of earlier literature reviews. The paper starts with a background where the field of CWM and other notations are defined. The scope of the review is given in Section 4. Thereafter numerical approaches for the coupled thermo-mechanical problem are briefly described followed by a section discussing modelling issues. Some recent published applications are reviewed in Section 8 and, finally, the future developments are discussed.

2. Conferences and reviews about computational welding mechanics

The two most important conferences in CWM are Mathematical Modelling of Weld Phenomena [6] and Trends in Welding Research [7]. The first series of conferences is devoted to simulations and associated experiments. The second series has a wider scope but with sessions focusing on CWM.

The book by Radaj [8] gives an overview of the different processes and phenomena that it is necessary to be aware of. Grøng [9] also discusses modelling with the focus on welding of aluminium. The three-part review by Lindgren [10–12] describes in detail different modelling techniques and review the publications in CWM up to 2001. The paper by Runesson et al. [13] is referred to for more about the thermo-dynamical basis for CWM. It also includes resistance spot welding.

* Tel.: +46 0920 913 06; fax: +46 0706 177 109.
E-mail address: lel@cad.luth.se

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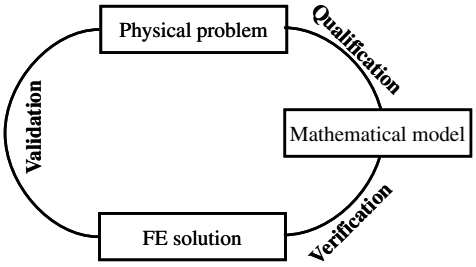


Fig. 1. Validation and verification in finite element modelling [14].

3. Definitions used in simulation and modelling

This paper is about simulations of welding processes using finite element models. The most important step in a simulation is to know why it is done. What is the purpose, *scope*? This determines what kind of model is needed and what accuracy is needed. The latter is also dependent on when the analysis is done, whether it is earlier or later in the development process. A *model*[1] is in our context a finite element model that will be used to represent certain aspects of the behaviour of a system. Those aspects that are of interest are the scope of the analysis. The definition of *simulation*[2] is an imitation of the internal processes and not merely the results of the system being simulated.

Validation [14], see Fig. 1, is the process where the accuracy of the model is evaluated by comparison with experimental results whereas *calibration* is the determination of parameters in order to create a match with some measurements. Thus the same measurements cannot be used both for calibration and validation purposes. An adequate model [15] is a model that has sufficient accuracy for its purpose. Two main difficulties in CWM that need both calibration and validation is the material model and heat input model. *Prediction* is the simulation of a specific case different from the validated case. *Verification* is the process where it is assured that the finite element model is correct with respect to the conceptual model and *qualification* is the same process but between conceptual model and reality [14]. The conceptual model is in our case the governing mathematical equations chosen when different elements, boundary conditions etc are defined in the finite element input file.

4. Scope of paper

The purpose of this review is to update the review by Lindgren [10–12] with recent publications and further extend the modelling practices given in [16]. The review is only discussing fusion welding processes. They are processes in which the metal parts are heated until they melt together. Thus we exclude e.g. explosive welding, friction welding, and spot welding. The welding can be performed with or without addition of filler material. Arc welding, electron beam welding and laser welding belong to this category of welding processes. The differences between these welding processes are expressed in terms of the distribution of heat and filler material in the thermo-mechanical model.

5. Numerical aspects of CWM

We discuss appropriate uncoupling of the general thermo-mechanical problem for CWM and describe the equations for thermal and mechanical analysis of problems with thermo(visco)plastic material behaviour and phase changes. The section ends with some remarks about meshing, adaptivity and parallel computing.

5.1. Couplings

The important couplings are shown in Fig. 2 and explained in Table 1. The fluid flow is not accounted for in an analysis of the mechanical effects of welding, with the exception of the work in Refs. [17–20]. This high temperature region is approximated by a soft solid in the mechanical analysis, a given heat input in the thermal analysis and sometimes a high conductivity for imitating convective heat transfer in the weld puddle. Thus the actual physics that takes place in the weld pool is simplified considerably and replaced by a *heat input model*. There exists complex models of these phenomena [21–23] but they have not been coupled to mechanical analyses. Advanced models can be used to predict the shape of the weld pool [22] and be used to refine the heat input models [24].

[1] http://www.hyperdictionary.com/search.aspx?Dict=&define=model.
[2] http://www.hyperdictionary.com/search.aspx?Dict=&define=model.

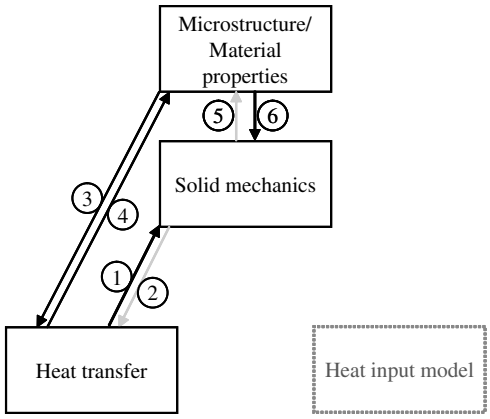


Fig. 2. Couplings in thermo-mechanical analysis of welding where a heat input model replaces the physics that takes place in the weld pool and its interaction with welding device. Less important couplings are grey.

Table 1
Couplings in Fig. 1

1	Thermal dilatation drives the deformation
2	Deformation generates a relatively small amount of heat in material due to plastic dissipation. Effect of elastic strain is even more negligible. The coupling exists also when the thermal boundary conditions are due to changing contact conditions with, for example, fixtures
3	Latent heat due to phase changes and thermal properties depend on microstructure and temperature
4	Temperature drives the microstructure changes
5	Deformation affects the microstructure changes
6	Mechanical properties depend on microstructure and temperature

Solving the thermal and mechanical fields simultaneously is not recommended as this means that a large un-symmetric system of equations must be solved, which is computationally demanding. A partitioning of the coupled fields by a staggered solution procedure [25] is more common to use to solve strongly coupled problems. This also simplifies the implementation as a modular approach can be used. The most common approach is a split of a time step into a pure thermal analysis followed by a mechanical analysis with the given temperatures. This approach is conditional stable but this is usually no problem in this kind of weakly coupled problems. However, there exists other splitting methods that are unconditional stable [26–29]. The inclusion of a microstructure model tightly coupled within the thermal analysis is called a thermo-metallurgical–mechanical (TMM) analysis. The effect of stress/strain on the microstructure is usually ignored for this type of welding simulations [30]. See Ronda and Oliver [31] where the latter coupling is discussed. They formulated a fully coupled problem without resorting to a staggered-approach and even derived a consistent tangent matrix for their TMM formulation. Couplings #2 and #5 are negligible and if the effect of temperature on microstructure, coupling #4, and thereby on the mechanical properties, coupling #6, can be predicted before the analysis, then it is possible to perform a weakly coupled analysis. Then the thermal analysis is completed first and the computed temperatures are used in the subsequent mechanical analysis.

The degree of the finite element shape functions for the displacements is usually one order higher than for the temperatures in order to have consistency between the thermal and mechanical fields. This is due to the fact that the temperature field directly becomes the thermal strain in the mechanical analysis. Usually the same mesh with linear elements is used in the thermal and mechanical analyses. Then the average temperature should be used for computing a constant thermal strain to be applied as the thermal load in the mechanical analysis. Oddy et al. [32] demonstrated the effect inconsistency between thermal and mechanical model on residual stresses. However, their proposal that this caused some discrepancies between measured and computed stresses in Lindgren and Karlsson [33] was not correct, see [10].

5.2. Reference frames

Two basic variants of reference frames have been used in CWM. The Updated Lagrangian (UL) approach is the most common but also Eulerian frames have been used to take advantage of the steady-state conditions with respect to the moving heat source that may exist. Eulerian analyses have been published in Refs. [34–45]. Computational savings from one to three orders of magnitude as compared to Lagrangian analyses are reported. However, these formulations are more complex to implement as the time derivatives is replaced by convective terms.

5.3. Thermal analysis

The standard *finite element semi-discretisation* procedure of the heat conduction leads to an energy relation [46,47]

$$\boldsymbol{M}\dot{\boldsymbol{H}} = \boldsymbol{F}_{\text{ext}} - \boldsymbol{F}_{\text{int}}, \tag{1}$$

where $\boldsymbol{M}$ is the mass matrix,[3] $\boldsymbol{F}_{\text{int}}$ is the internal flux vector and $\boldsymbol{F}_{\text{ext}}$ is the thermal load vector. The latter may include plastic dissipated heat and the effect of elastic strains from the mechanical analysis in a strongly coupled analysis. This set can be solved in terms of enthalpies or temperatures as we have the enthalpy per unit mass

$$H(T) = \int_{T_{\text{ref}}}^{\text{T}} c_{\text{p}}(\tau)\,\mathrm{d}\tau + \text{latent heats}, \tag{2}$$

where c_p is the heat capacity at constant pressure. The difference between c_p and c_v, the heat capacity at constant strain is negligible for solids. The relation between enthalpy and temperature above leads to a temperature based formulation

$$\boldsymbol{C}\dot{\boldsymbol{T}} = \boldsymbol{F}_{\text{ext}} - \boldsymbol{F}_{\text{int}}, \tag{3}$$

where $\boldsymbol{C}$ is the heat capacity matrix.

The *spatial discretisation* leading to Eq. (3) is based on the finite element interpolation of the temperature field within each element by

$$T(\boldsymbol{x}, t) = \boldsymbol{N}(\boldsymbol{x})\boldsymbol{T}^{\text{e}}(t), \tag{4}$$

where $\boldsymbol{x}$ is the current coordinates, $\boldsymbol{N}(\boldsymbol{x})$ is a matrix with the interpolation (shape) functions and $\boldsymbol{T}^{\text{e}}(t)$ is the element temperature vector. The geometry and the mechanical variables are assumed to be fixed during the increment (rigid conductor). Thus the current coordinates, $\boldsymbol{x}$, are the coordinates at the beginning of a time step. The gradient of the temperature field is

$$\boldsymbol{g} = \nabla\boldsymbol{T} = \nabla\boldsymbol{N}(\boldsymbol{x})\boldsymbol{T}^{\text{e}}(t) = \boldsymbol{B}(\boldsymbol{x})\boldsymbol{T}^{\text{e}}(t), \tag{5}$$

where the matrix $\boldsymbol{B}$ has the first derivatives of the interpolation functions w.r.t. coordinates.

Fourier's equation for the internal heat flux is

$$\boldsymbol{q}_{\text{int}} = -\boldsymbol{\lambda}\boldsymbol{g}, \tag{6}$$

where $\boldsymbol{\lambda}$ is a diagonal conductivity matrix.

The element matrices $\boldsymbol{N}$, $\boldsymbol{B}$, and $\boldsymbol{\lambda}$ and the density ρ and heat capacity c are used to compute element contributions to Eq. (3). We have

$$\boldsymbol{c} = \int_{v^{\text{e}}} \rho c(T)\boldsymbol{N}^{\text{T}}\boldsymbol{N}\,\mathrm{d}v \overset{\text{assembling}}{\longrightarrow} \boldsymbol{C} \tag{7}$$

and

$$\boldsymbol{f}_{\text{int}} = -\int_{v^{\text{e}}} \boldsymbol{B}^{\text{T}}\boldsymbol{q}_{\text{int}}\,\mathrm{d}v = \int_{v^{\text{e}}} \boldsymbol{B}^{\text{T}}\boldsymbol{\lambda}\boldsymbol{B}\,\mathrm{d}v\boldsymbol{T}^{\text{e}} = \boldsymbol{k}^{\text{e}}\boldsymbol{T}^{\text{e}} \overset{\text{assembling}}{\longrightarrow} \boldsymbol{F}_{\text{int}} = \boldsymbol{K}T, \tag{8}$$

where v^{e} is the current volume of the element and $\boldsymbol{k}^{\text{e}}$ is the temperature dependent element conductivity matrix.

The load vector $\boldsymbol{F}_{\text{ext}}$ can have element contributions due to volumetric heating as well as from nodal or surface heat flux. Boundary conditions that are dependent on the temperature of the body, like convection and radiation, will also contribute to the conductivity matrix.

The *temporal discretisation* is often based on the generalised midpoint method. It aims at fulfilling the energy equilibrium at time

$${}^{n+\theta}t = {}^{n}t + \theta\Delta t, \tag{9}$$

where n is the increment/time step number, ${}^{n}t$ is the time at the start of the time step, Δt is the length of the current time step and the parameter $\theta \in [0,1]$ determines where in the increment we want to fulfil the energy equilibrium. The method is unconditional stable for if $\theta \geqslant 0.5$. The use of 0.5 gives a second order accurate time stepping but the solution can show oscillations. Further details about stability of the time stepping can be found in [46,48,49]. The author prefers to use as stable approach as possible, i.e. $\theta = 1$ and combined with a lumped heat capacity matrix that can improve the solution in the transition region between the large temperature gradient near the arc and the surrounding low temperature region.

[3] Bold face denotes vectors and matrices.

The use of a consistent heat capacity matrix may give negative temperatures near the weld arc due to "overshooting" and this is nearly eliminated when using a lumped matrix.

The nonlinearities in Eq. (3) require an iterative approach for solving the temperature at each time step [46,50]. A *predictor* is used for the first iteration and thereafter an *iterative corrector* phase is needed. The predictor can be as simple as assuming no change in temperature during the time step. The corrector phase aims at fulfilling Eq. (3) at time ${}^{n+\theta}t$. We rewrite the equation for this purpose to

$$ {}^{n+\theta}_{i+1}\boldsymbol{R} = \left[{}^{n+\theta}_{i+1}\boldsymbol{F}_{\text{ext}} - {}^{n+\theta}_{i+1}\boldsymbol{F}_{\text{int}} - {}^{n+\theta}_{i+1}\boldsymbol{C}\, {}^{n+\theta}_{i+1}\dot{\boldsymbol{T}} \right] \Delta t, \tag{10} $$

where the residual $\boldsymbol{R}$ should become zero and the left subscript is an iteration counter.

The midpoint temperature is

$$ {}^{n+\theta}_{i}\boldsymbol{T} = {}^{n}\boldsymbol{T} + \theta\, {}_{i}\Delta\boldsymbol{T} \tag{11} $$

and we introduce iterative corrections ${}_{i}\delta\boldsymbol{T}$ to the increment in temperature

$$ {}_{i+1}\Delta\boldsymbol{T} = {}_{i}\Delta\boldsymbol{T} + {}_{i}\delta\boldsymbol{T}. \tag{12} $$

The midpoint method uses

$$ {}^{n+\theta}_{i+1}\dot{\boldsymbol{T}} = \frac{{}^{i+1}\Delta\boldsymbol{T}}{\Delta t} = {}^{n+\theta}_{i}\dot{\boldsymbol{T}} + \frac{{}_{i}\delta\boldsymbol{T}}{\Delta t}. \tag{13} $$

The *Newton–Raphson* iterative approach is based on a Taylor expansion of Eq. (10) where we want to find the iterative correction that may give ${}^{n+\theta}_{i+1}\boldsymbol{R} = \boldsymbol{0}$ in the next iteration.

$$ {}^{n+\theta}_{i+1}\boldsymbol{R} \approx {}^{n+\theta}_{i}\boldsymbol{R} + \frac{\partial^{n+\theta}_{i}\boldsymbol{R}}{\partial\boldsymbol{T}}\, {}_{i}\delta\boldsymbol{T} = \boldsymbol{0}. \tag{14} $$

This can be written as

$$ -\frac{\partial^{n+\theta}_{i}\boldsymbol{R}}{\partial\boldsymbol{T}}\, {}_{i}\delta\boldsymbol{T} = {}^{n+\theta}_{i}\boldsymbol{R} \tag{15} $$

or introducing the tangent matrix

$$ {}^{n+\theta}_{i}\boldsymbol{K}_{\text{t}}\, {}_{i}\delta\boldsymbol{T} = {}^{n+\theta}_{i}\boldsymbol{R}. \tag{16} $$

The tangent matrix needed to retain second order convergence of the Newton–Raphson method in this iterative process is

$$ {}^{n+\theta}_{i}\boldsymbol{K}_{\text{t}} = -\frac{\partial^{n+\theta}_{i}\boldsymbol{R}}{\partial\boldsymbol{T}} = \left[\frac{\partial\left({}^{n+\theta}_{i}\boldsymbol{C}\, {}^{n+\theta}_{i}\dot{\boldsymbol{T}}\right)}{\partial\boldsymbol{T}} + \frac{\partial^{n+\theta}_{i}\boldsymbol{F}_{\text{int}}}{\partial\boldsymbol{T}} - \frac{\partial^{n+\theta}_{i}\boldsymbol{F}_{\text{ext}}}{\partial\boldsymbol{T}} \right] \Delta t. $$

It is usually approximated to

$$ {}^{n+\theta}_{i}\boldsymbol{K}_{\text{t}} \approx \left[{}^{n+\theta}_{i}\boldsymbol{C}\, \frac{1}{\Delta t} + {}^{n+\theta}_{i}\boldsymbol{K}\theta \right] \Delta t = {}^{n+\theta}_{i}\boldsymbol{C} + \theta\Delta t\, {}^{n+\theta}_{i}\boldsymbol{K}. \tag{17} $$

A modified Newton–Raphson approach may be computationally more efficient. Then the tangent matrix is not updated every iteration. Typically it is updated during the first 1–2 iterations and thereafter only if divergence occurs. The incremental, iterative approach for the nonlinear heat conduction problem is summarised in Fig. 3.

The temperature formulation, Eq. (3), is the most common used formulation for computing the thermal fields in welding. However, the enthalpy formulation, Eq. (1), has less numerical problems in the presence of latent heats and therefore was used in [5,51,52].

Another numerical approach for relieving the numerical problems due to latent heats is the fictitious heat source method [53] used by Murty et al. [54]. A third approach, used by e.g. Lindgren et al. [16] and many others is based on a time-averaging procedure for the heat capacity [46,55]. It is sometimes also called an enthalpy method. The heat capacity is then taken as

$$ {}^{n+\theta}c = c_{\text{eff}} = \frac{{}^{n+1}_{i}H - {}^{n}H}{{}_{i}\Delta T}, \quad \text{if } {}_{i}\Delta T \neq 0 \tag{18} $$

or, in case of severe convergence problems, the average is computed from previous time step

$$ {}^{n+\theta}c = c_{\text{eff}} = \frac{{}^{n}H - {}^{n-1}H}{{}^{n}T - {}^{n-1}T}. \tag{19} $$

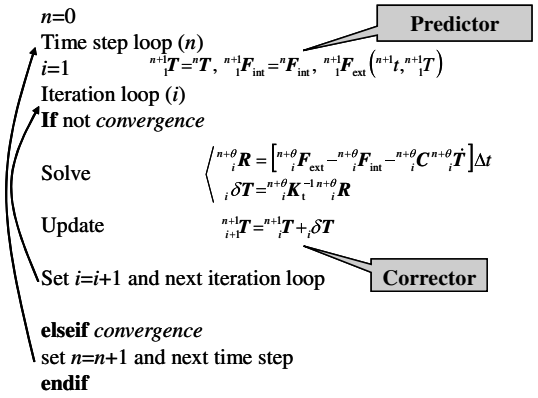


Fig. 3. Incremental, iterative procedure for thermal analysis. Initial conditions give ${}^0T$.

The enthalpy method, Eq. (1), works even for a pure metal. Other averaging methods work well for alloys where the melting occurs over a temperature range. Pham [56] compared space-averaging with space-averaging techniques for regularising the latent heat effect and also discussed the use of lumped heat capacity matrix. The most straightforward way to reduce the nonlinearity is to extend the temperature range over which the latent heat is released/consumed in the model.

5.4. Mechanical analysis

A quasistatic mechanical analysis is sufficient as the inertia effects in fusion welding processes are negligible as the weld pool is simplified considerably as described in Section 6.3. Welding deformations may quite often be visible to the naked eye implying that large deformation and rotation effects are important but the strains are usually small. The latter is also due to the simplification of the weld pool region. However, the additional cost of accounting for large strains is small. Therefore accounting for large deformations makes the mechanical analysis more complex than the thermal analysis and there exists several options for how to treat them. It is possible to use different mesh descriptions, different kinetic and kinematic variables, and constitutive models, see [47,50,57–59]. The most common approach is described in this paper. It is based on the *Updated Lagrangian* (UL) formulation using the *Cauchy* (*true*) *stress* and the *rate of deformation tensor* (*velocity strain*) as conjugate stress and strain measures. The latter corresponds to the use of the logarithmic (natural) strain definition. The stress–strain relations can be based on a hyperelastic model or *hypoelastic* model. The latter is the more common approach. The hypoelastic models do not exactly correspond to an elastic material but the error disappears in the case of metal plasticity as the elastic strains are small for this case [60]. The use of hypoelastic relations has also the drawback that one must choose an objective stress rate measure. The Jaumann rate was popular initially but was replaced by the *Green–Naghdi stress rate* as it has less problems with oscillations for large shear deformation cases [58,61].

The displacement based, finite element semi-discretisation [47,57] leads to

$$\boldsymbol{M}\ddot{\boldsymbol{U}} \approx \boldsymbol{0} = \boldsymbol{F}_{\text{ext}} - \boldsymbol{F}_{\text{int}}, \tag{20}$$

where $\boldsymbol{M}$ is the (ignored) mass matrix, $\boldsymbol{F}_{\text{int}}$ is the internal forces due to the stresses and $\boldsymbol{F}_{\text{ext}}$ is the external loads. The nonlinearities in Eq. (20) require an incremental, iterative approach for solving the displacements at each time step [50,58]. It is recommended to combine it with line search [50] if no contact surfaces are present in the model. Its application in the context of contact needs some modification of the algorithm and its effectiveness is reduced [50].

The standard finite element formulation uses only interpolation of displacement fields. They are interpolated over each element as

$${}_i^{n+1}\boldsymbol{u}\left({}_i^{n+1}\boldsymbol{x}, t\right) = \boldsymbol{N}\left({}_i^{n+1}\boldsymbol{x}\right){}_i^{n+1}\boldsymbol{u}(t), \tag{21}$$

where $\boldsymbol{N}\left({}_i^{n+1}\boldsymbol{x}\right)$ is a matrix with the interpolation (shape) functions and ${}_i^{n+1}\boldsymbol{u}$ is the element displacement vector. The coordinate ${}_i^{n+1}\boldsymbol{x}$ is the current estimate of the coordinate at the end of the time step and therefore we have added a left subscript as an iteration counter.

The rate of deformation tensor (velocity strain) is computed as

$${}_i^{n+1}\boldsymbol{d} = \frac{1}{2}\left(\nabla_i^{n+1}\dot{\boldsymbol{u}}^{\mathrm{T}} + \nabla_i^{n+1}\dot{\boldsymbol{u}}\right) = {}_i^{n+1}\boldsymbol{B}_i^{n+1}\dot{\boldsymbol{u}}, \tag{22}$$

where the ${}_i^{n+1}\boldsymbol{B}$-matrix relates velocity with velocity strain. The time integral of Eq. (22) over a time step gives the increment in strain which give increment in stresses. The stress–strain relations are discussed briefly in Section 5.4.1. The internal forces are obtained from stresses as

$$ {}^{n+1}_{i}\boldsymbol{f}_{\text{int}} = \int_{{}^{n+1}_{i}v^{\text{e}}} {}^{n+1}_{i}\boldsymbol{B}^{\text{T}}\, {}^{n+1}_{i}\boldsymbol{\sigma}\, \mathrm{d}v \overset{\text{assembling}}{\rightarrow} {}^{n+1}_{i}\boldsymbol{F}_{\text{int}}, \tag{23} $$

where ${}^{n+1}_{i}\boldsymbol{\sigma}$ is the Cauchy stress (true stress) in vector form and the integration is taken over the current volume of the element. ${}_{i}{}^{n+1}\boldsymbol{F}_{\text{ext}}$ gets contributions from element volume forces as well as from boundary conditions. Note that the load due to the thermal strain is assembled to the internal force vector via the constitutive model.

The quasistatic formulation does not need any temporal discretisation.

The updating of the current geometry is

$$ {}^{n+1}_{i}\boldsymbol{x} = {}^{n}\boldsymbol{x} + {}_{i}\Delta\boldsymbol{x} = {}^{n}\boldsymbol{x} + {}_{i}\Delta\boldsymbol{x}, \tag{24} $$

where the change in coordinate is the incremental displacement. The iterative procedure will give corrections to this increment as

$$ {}_{i+1}\Delta\boldsymbol{x} = {}_{i}\Delta\boldsymbol{x} + {}_{i}\delta\boldsymbol{x}. \tag{25} $$

Equilibrium at the end of the time step/increment is obtained when the residual $\boldsymbol{R}$ is zero

$$ {}^{n+1}_{i}\boldsymbol{R} = {}^{n+1}_{i}\boldsymbol{F}_{\text{ext}} - {}^{n+1}_{i}\boldsymbol{F}_{\text{int}} = \boldsymbol{0}. \tag{26} $$

The *Newton–Raphson* iterative approach is based on a Taylor expansion of Eq. (26) where we want to find the iterative correction that may give ${}^{n+1}_{i+1}\boldsymbol{R} = \boldsymbol{0}$ in the next iteration.

$$ {}^{n+1}_{i+1}\boldsymbol{R} \approx {}^{n+1}_{i}\boldsymbol{R} + \frac{\partial^{n+1}_{i}\boldsymbol{R}}{\partial\boldsymbol{x}}\,{}_{i}\delta\boldsymbol{x} = \boldsymbol{0}. \tag{27} $$

This can be written as

$$ -\frac{\partial^{n+1}_{i}\boldsymbol{R}}{\partial\boldsymbol{x}}\,{}_{i}\delta\boldsymbol{x} = {}^{n+1}_{i}\boldsymbol{R} \tag{28} $$

or introducing the tangent matrix

$$ {}^{n+1}_{i}\boldsymbol{K}_{\text{t}}\,{}_{i}\delta\boldsymbol{x} = {}^{n+1}_{i}\boldsymbol{R}. \tag{29} $$

The tangent matrix is

$$ {}^{n+1}_{i}\boldsymbol{K}_{\text{t}} = -\frac{\partial^{n+1}_{i}\boldsymbol{R}}{\partial\boldsymbol{x}} = \frac{\partial^{n+1}_{i}\boldsymbol{F}_{\text{int}}}{\partial\boldsymbol{x}} - \frac{\partial^{n+1}_{i}\boldsymbol{F}_{\text{ext}}}{\partial\boldsymbol{x}}. \tag{30} $$

The dependency of external forces on displacement is often ignored giving

$$ {}^{n+1}_{i}\boldsymbol{K}_{\text{t}} \approx \frac{\partial^{n+1}_{i}\boldsymbol{F}_{\text{int}}}{\partial\boldsymbol{x}}. \tag{31} $$

Furthermore, the internal forces have several contributions that depend on the deformation. The domain of the integral in Eq. (23), the ${}^{n+1}_{i}\boldsymbol{B}_{\text{int}}$-matrix and the stresses depend on the deformation. Usually, only the dependency of the stresses on the displacements is included leading to

$$ {}^{n+1}_{i}\boldsymbol{k}_{\text{t}} \approx \frac{\partial^{n+1}_{i}\boldsymbol{f}_{\text{int}}}{\partial\boldsymbol{u}} = \frac{\partial}{\partial\boldsymbol{u}}\left[\int_{{}^{n+1}_{i}v^{\text{e}}} {}^{n+1}_{i}\boldsymbol{B}^{\text{T}}\, {}^{n+1}_{i}\boldsymbol{\sigma}\, \mathrm{d}v\right] \approx \int_{{}^{n+1}_{i}v^{\text{e}}} {}^{n+1}_{i}\boldsymbol{B}^{\text{T}}\, \frac{\partial^{n+1}_{i}\boldsymbol{\sigma}}{\partial\boldsymbol{u}}\, \mathrm{d}v, \tag{32} $$

where we have used the element displacement vector $\boldsymbol{u}$ instead of the global displacement vector $\Delta\boldsymbol{x}$ in the notations. The derivative of the stresses with respect to the displacement can be written as

$$ {}^{n+1}_{i}\boldsymbol{k}_{\text{t}} \approx \int_{{}^{n+1}_{i}v^{\text{e}}} {}^{n+1}_{i}\boldsymbol{B}^{\text{T}}\, \frac{\partial^{n+1}_{i}\boldsymbol{\sigma}}{\partial\boldsymbol{\varepsilon}}\, \frac{\partial\boldsymbol{\varepsilon}}{\partial\boldsymbol{u}}\, \mathrm{d}v = \int_{{}^{n+1}_{i}v^{\text{e}}} {}^{n+1}_{i}\boldsymbol{B}^{\text{T}}\, {}^{n+1}_{i}\boldsymbol{c}_{\text{t}}\, {}^{n+1}_{i}\boldsymbol{B}\, \mathrm{d}v, \tag{33} $$

where we have excluded the contribution to stress changes due to rigid body rotation in the last relation above. See Refs. [57–59] for details about the geometric stiffness matrix that accounts for this effect. The derivative of stresses with respect to strain

$$ {}^{n+1}_{i}\boldsymbol{c}_{\text{t}} = \frac{\partial^{n+1}_{i}\boldsymbol{\sigma}}{\partial\boldsymbol{\varepsilon}} \tag{34} $$

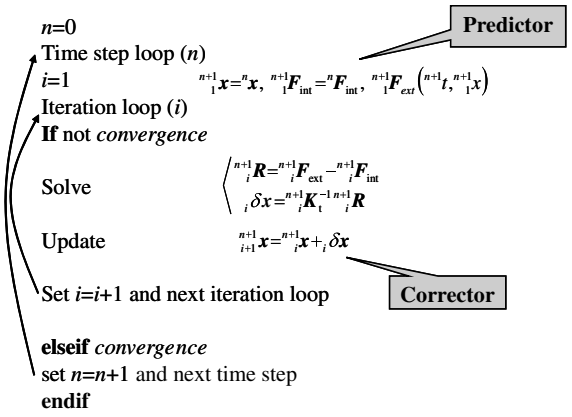


Fig. 4. Incremental, iterative procedure for mechanical analysis. Initial conditions give ${}^0x$.

depends on the computational algorithm for the stress updating algorithm and is called a consistent or algorithmic constitutive matrix [62–64].

Thus we have

$$ {}^{n+1}_{i}\boldsymbol{k}_{\mathrm{t}} = \int_{{}^{n+1}_{i}v^{\mathrm{e}}} {}^{n+1}_{i}\boldsymbol{B}^{\mathrm{T}}\, {}^{n+1}_{i}\boldsymbol{c}_{\mathrm{t}}\, {}^{n+1}_{i}\boldsymbol{B}\,\mathrm{d}v \overset{\text{assembling}}{\rightarrow} {}^{n+1}_{i}\boldsymbol{K}_{\mathrm{t}}. \tag{35} $$

The incremental, iterative approach for the nonlinear deformation problems is summarised in Fig. 4.

Linear elements, interpolating functions, are usually preferred in analysis of problems in plasticity. Linear quad, in 2D, and brick, in 3D, elements are the basic recommendation in plasticity [65,66]. They perform better than displacement based linear triangles or tetrahedrons. The linear elements require reduced integration of the volumetric strain or enhanced strain fields [67] in order to avoid locking in case of large plastic strains due to the plastic incompressibility constraint. See Belytschko et al. [57] for a discussion of different elements and locking due to plastic incompressibility. The flexibility in mesh generation for triangle and tetrahedron elements is the motivation for continuing developments of this type of elements using separate interpolations for pressure and displacements [68].

The stress updating is described next. The focus is on the use of an unrotated configuration, given in Section 5.4.1, as a basis for the hypoelastic stress update of the constitutive model, with some examples in Section 5.4.2. Finally, the stress–strain algorithm is described in Section 5.4.3.

5.4.1. Unrotated configuration and hypoelastic stress–strain relations

The hypoelastic formulation relates increment in strain to increment in stress. These increments must be based on objective rates of stress and strain where the effect of a rigid body rotation is excluded.

The spatial velocity gradient, Eq. (22), is an objective quantity [59]. Hughes and Winget [69] showed that the formula below is a second order accurate approximation of the integral of the velocity gradient to give a strain increment

$$ {}_{i}\Delta\boldsymbol{\varepsilon} = \int_{{}^{n}t}^{{}^{n+1}t} {}_{i}\boldsymbol{d}(\tau)\,\mathrm{d}\tau \approx {}^{n+1/2}_{i}\boldsymbol{B}_{i}\Delta\boldsymbol{u}, \tag{36} $$

where ${}^{n+1/2}_{i}\boldsymbol{B}$ means that the matrix with derivatives of the shape functions is evaluated for the current estimate of the midpoint geometry ${}^{n+1/2}_{i}\boldsymbol{x}$. An unrotated stress, ${}^{n}\boldsymbol{\sigma}_{\mathrm{R}}$, at time ${}^{n}t$ is created by

$$ {}^{n}\boldsymbol{\sigma}_{\mathrm{R}} = {}^{n}\boldsymbol{R}^{\mathrm{T}\,n}\boldsymbol{\sigma}. \tag{37} $$

Note that $\boldsymbol{R}$ is the rotation matrix and $\boldsymbol{\sigma}$ is the stress vector. The equation is not in tensor form. The stress, backstress (${}^{n+1}\boldsymbol{\alpha}$) and plastic strain variables (${}^{n+1}\boldsymbol{\varepsilon}^{\mathrm{p}}$) at state ${}^{n}t$ are all unrotated using ${}^{n}\boldsymbol{R}$ and the strain increment that is associated to the midpoint geometry is unrotated using ${}^{n+1/2}_{i}\boldsymbol{R}$. Thereafter the stress updating is done in this unrotated configuration. This is based on a chosen constitutive model, Section 5.4.2, and stress updating algorithm, Section 5.4.3. The stress–strain algorithm can be written symbolically as

$$ \left({}^{n+1}\boldsymbol{\sigma}_{\mathrm{R}}, {}^{n+1}\boldsymbol{\alpha}_{\mathrm{R}}, {}^{n+1}\boldsymbol{\varepsilon}^{\mathrm{p}}_{\mathrm{R}}\right) = \text{stress-algorithm}({}_{i}\Delta\boldsymbol{\varepsilon}_{\mathrm{R}}, {}^{n}\boldsymbol{\sigma}_{\mathrm{R}}, {}^{n}\boldsymbol{\alpha}_{\mathrm{R}}, {}^{n}\boldsymbol{\varepsilon}^{\mathrm{p}}_{\mathrm{R}}, {}^{n}T, \Delta T, \ldots). \tag{38} $$

The updated variables must thereafter also be rotated forward to time ${}^{n+1}t$. The use of a rotation matrix $\boldsymbol{R}$ from the polar decomposition of the deformation gradient in the relations above corresponds to the use of the Green–Naghdi, $\dot{\boldsymbol{\sigma}}_{\mathrm{GN}}$, objective stress rate [58].

5.4.2. Constitutive models

Choosing material model and obtaining material parameters are crucial tasks in CWM. Nearly all models that have been used in welding simulations are deviatoric plasticity models. Two types of models have been used; rate-independent (plastic) and rate-dependent (viscoplastic) models. Phase changes effects and other special considerations in CWM are discussed in the modelling section and are not included below. More details and examples of models and properties are found in [11].

An *additive decomposition of the strain rates* is assumed. This can be motivated in different ways starting from the multiplicative decomposition of the deformation gradient, see for example [58,62,70–72]. Some of the differences in these derivations disappear when assuming small elastic strains [58,60]. This is reasonable for metal plasticity. Thus it is assumed[4]

$$\dot{\boldsymbol{\varepsilon}} = \dot{\boldsymbol{\varepsilon}}^{\mathrm{e}} + \dot{\boldsymbol{\varepsilon}}^{\mathrm{p}} + \dot{\boldsymbol{\varepsilon}}^{\mathrm{th}}, \tag{39}$$

where $\dot{\boldsymbol{\varepsilon}}$ is the total strain rate, $\dot{\boldsymbol{\varepsilon}}^{\mathrm{e}}$ is the elastic strain rate, $\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}}$ is the plastic strain rate and $\dot{\boldsymbol{\varepsilon}}^{\mathrm{th}}$ rate of the thermal strain.

The *elastic behaviour* is usually described by a hypoelastic relation [58]

$$\hat{\boldsymbol{\sigma}}_{\mathrm{GN}} = \boldsymbol{E}^{\mathrm{e}}\dot{\boldsymbol{\varepsilon}}^{\mathrm{e}} + \frac{\mathrm{d}\boldsymbol{E}^{\mathrm{e}}}{\mathrm{d}T}\dot{T}\boldsymbol{\varepsilon}^{\mathrm{e}}, \tag{40}$$

where $\dot{\boldsymbol{\sigma}}_{\mathrm{GN}}$ is an objective stress rate (here the Green–Naghdi rate) and E^{e} is a matrix with the elastic material properties.

A surface, Φ, is used to define elastic respectively plastic stress processes. A stress state inside, $\Phi \leqslant 0$, or on the surface $\Phi = 0$ with $\dot{\Phi} \leqslant 0$, corresponds to an elastic deformation process. The yield surface is defined as

$$\Phi = \bar{\sigma} - \sigma_{\mathrm{y}}, \tag{41}$$

where σ_{y} is called the yield limit and $\bar{\sigma}$ is the effective von Mises stress

$$\bar{\sigma} = \sqrt{\frac{3}{2}\boldsymbol{\xi}^{\mathrm{T}}\boldsymbol{P}_{\mathrm{M}}\boldsymbol{\xi}}, \tag{42}$$

where

$$\boldsymbol{\xi} = \boldsymbol{s} - \boldsymbol{\alpha} \tag{43}$$

is the deviatoric stress, $\boldsymbol{s}$, minus the backstress $\boldsymbol{\alpha}$. $\boldsymbol{P}_{\mathrm{M}}$ is a matrix combining the projection of stresses to the deviatoric space, the use of vector notation and the definition of the effective stress. Ref. [73] shows how it is used for general as well as plane stress states. Crisfield [58] describes how it can also accommodate anisotropic plasticity.

Furthermore, maximum plastic dissipation is postulated leading to *associated plasticity*. The yield surface is the plastic potential for the plastic strains. It gives the components of the rate of the plastic strains as

$$\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}} = \boldsymbol{L}\dot{\lambda}\frac{\partial \Phi}{\partial \boldsymbol{\sigma}} = \dot{\bar{\varepsilon}}^{\mathrm{p}}\frac{3}{2\bar{\sigma}}\boldsymbol{L}\boldsymbol{\xi}, \tag{44}$$

where $\boldsymbol{L}$ is a diagonal matrix with the diagonal

$$\boldsymbol{L}_{\mathrm{diag}} = [\,1 \quad 1 \quad 1 \quad 2 \quad 2 \quad 2\,]. \tag{45}$$

The effective plastic strain rate $\dot{\bar{\varepsilon}}^{\mathrm{p}}$ is used as a measure of the magnitude of the plastic strain rates. It is defined by $\bar{\sigma}\dot{\bar{\varepsilon}}^{\mathrm{p}} = \boldsymbol{\sigma}^{\mathrm{T}}\dot{\boldsymbol{\varepsilon}}^{p}$ leading to

$$\dot{\bar{\varepsilon}}^{\mathrm{p}} = \sqrt{\frac{2}{3}\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}^{\mathrm{T}}}\boldsymbol{L}^{-1}\dot{\boldsymbol{\varepsilon}}^{\mathrm{p}}}. \tag{46}$$

The magnitude of the effective plastic strain rate is obtained from the so-called *consistency condition in the case of rate-independent plasticity*

$$\dot{\Phi} \equiv 0 \tag{47}$$

as we can never have a stress state outside the yield surface in this model.

The magnitude of the effective plastic strain rate is obtained from *the flow strength equation in the case of rate-dependent plasticity*. Relations where the rate is a monotonous increasing function of the excess stress [74] or overstress, $\Phi > 0$, used in

$$\dot{\bar{\varepsilon}}^{\mathrm{p}} = \eta(\gamma \cdot \phi\langle\Phi\rangle), \tag{48}$$

where the bracket $\langle\rangle$ denotes that the rate is zero if $\Phi \leqslant 0$. The introduction of γ and ϕ inside η facilitates the identification with different models. See [74–79] for further details.

[4] We use the so-called Voigt notation and engineering strain definition.

The *hardening/softening* behaviour of the material can have an *isotropic* and a *kinematic* part. *The isotropic hardening* can include recovery processes. They may be accommodated in the yield limit, σ_y, but also in the viscosity, γ, of the flow strength equation, Eq. (48). The *kinematic hardening* is accounted for by translating the yield surface by changing the back-stress $\boldsymbol{\alpha}$. The evolution equations for the hardening can also include static and dynamic recovery processes [80].

5.4.3. Stress updating algorithm

The stress–strain algorithm has two tasks. They are; computation of stress from given strain and computing the consistent constitutive matrix. The first task must be done efficiently and accurately and will be described briefly and further details are in the references. The computation of a consistent tangent is required when the second order convergence of the Newton–Raphson method should be preserved. The latter can be approximated at the cost of slower convergence in the full Newton–Raphson method. The computation of the consistent constitutive matrix depends on the material model and the numerical algorithm used [62,63]. The stress updating algorithm, symbolically in Eq. (38), will be form-identical with the small strain logic when it is formulated using the concepts and models as described in [81]. Crisfield [58] shows that this is the case only for cases where a symmetric tensor for the direction of plastic flow cancel the antisymmetric rotation terms in the stress rate in the consistency condition as in the case of associated plasticity. An operator-split approach where an elastic predictor is followed by a plastic corrector can be used. The stress updating for the rate-dependent and rate-independent plasticity models can be formulated in similar ways where the first is a sub case of the latter [57,58,64,82–85]. A logarithmic transformation of the flow strength equation can be applied if it is too stiff so that the Newton–Raphson scheme cannot find the increment in effective plastic strain [86].

5.5. Parallel computing

Parallel computing makes it possible to reduce the wall clock time for an analysis. Usually, implicit finite element methods are used in welding simulations and these are less scalable for parallel computing than explicit finite element methods. However, sometimes the limiting factor is computer memory and then it is possible to obtain a near linear gain with the number of processors. Lenz and Rick [87] was able to reduce the wall clock time by more than a factor of two when going from one to two processors using Marc. Ericsson et al. [88][5] also analysed a large structure using Marc and parallel computing. Initial evaluations, not shown in their paper, using large models with solid elements gave a speed up of 1.7 when going from four to eight processors.

5.6. Dynamic and adaptive meshing

Dynamic and adaptive meshing are methods to concentrate the elements to regions with large gradients. Thus, increased computational efficiency can be obtained without sacrificing accuracy. Dynamic meshing denotes a technique where the user prescribed the refinement and coarsening of the finite element mesh. Typically a fine mesh region is moving with the heat source [89,90]. Lindgren et al. [89] reduced the computer time by a factor of two using dynamic meshing. Lenz and Rick [87] only obtained a 20% reduction using dynamic meshing in the code Marc. Runnemalm and Hyun [91] decreased the computational time by a factor of two using adaptive meshing and Hemmer et al. [92] reduced the time by a factor of 2–4.

The use of error measures to guide the mesh refinement and coarsening is a more accurate approach than using dynamic meshing. Runnemalm and Hyun [91] found that it was advisable to combine thermal and mechanical measures when determining mesh refinement or coarsening and that the pattern for remeshing is not so easy to predict without an error measure. Rieger and Wriggers [93] discussed adaptive meshing using error estimators for coupled, nonlinear elastic problems where different meshes are used in thermal and mechanical analyses. Adaptive meshing can also be combined with distortion metric to avoid too distorted elements. This can be particular useful when simulating multipass welding [94]. Different error measures and adaptive strategies are discussed in [95–98] and the Oddy distortion metric [99] is useful for indicating where the mesh is distorted.

5.7. Substructuring

Substructuring is an efficient tool for reducing the size of large models for linear, elastic analyses. Usually some approximations are introduced when implemented for non-linear problems. The basic idea is to only include the region around the weld in the system of equations. The rest of the structure will be assumed to behave linearly and condensed out. Thus one must ignore the temperature dependency of the material properties and the large deformations in the condensed part of the

[5] Private communication.

structure. This can be done by setting the boundary for this region where the temperature only will reach about 200–300 °C. This will not reduce the size of the problem so much as most elements are near the weld. Therefore a dynamic substructuring is needed that is moving with the heat source. It is also necessary to obtain the thermal load from the condensed part to the analysed part. The thermal load cannot be obtained without doing a recovery step for obtaining deformations, stresses and thermal loads on this part and then some of the gain in speed is lost.

Fuglsang Andersen [100] combined substructuring and dynamic remeshing. A ship subassembly was modelled by linear, elastic shell elements. This global model was the basic information carrying model, Fig. 5a. The panel had eight weld lines, as each stiffener was welded from both sides. A local solid model was created of the region whose welding was going to be simulated, see Fig. 5b. Dynamic remeshing was used for the local model. The stiffness of the global model was increased by each weld pass, as there were initially only tack welds connecting web and plate. The tack welds corresponded to some shell elements connections. The result from each local model was mapped to the global model before going to the next weld pass. The assembly required eight local models. The substructuring was not applied to the thermal model and there was no interaction between the thermal models for each weld submodel, as it was assumed that it had cooled to room temperature before analyzing next local model. Fuglsang Andersen [100] was able to simulate a total of 8 m welding and studied the effect of the welding sequence on the residual distortion of the panel, see Fig. 6.

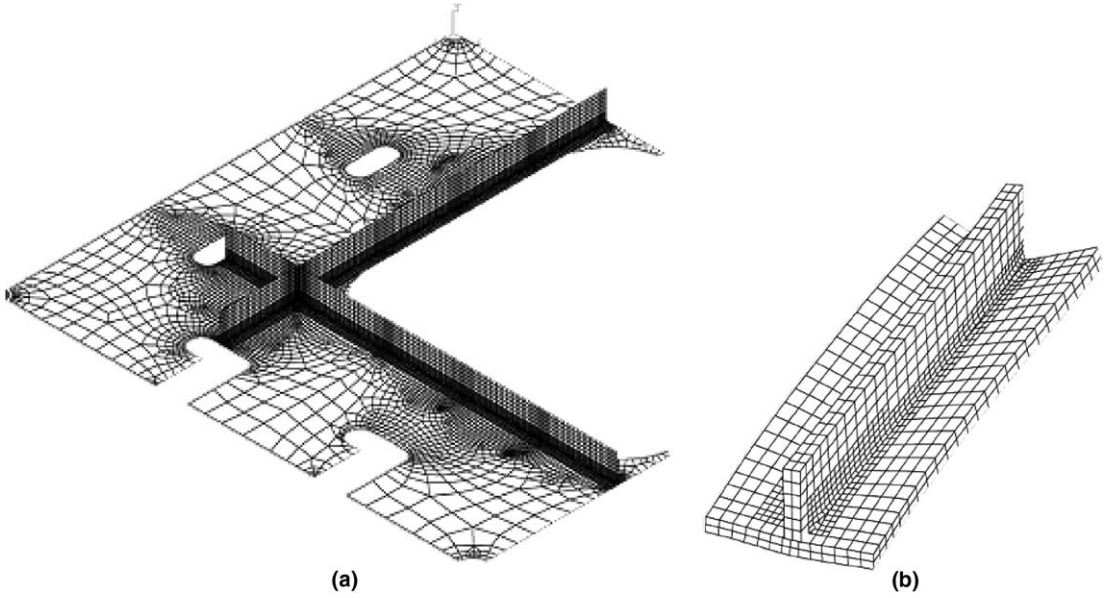


Fig. 5. Global, shell model of ship subassembly (a) and solid model of part to be welded (b) [100].

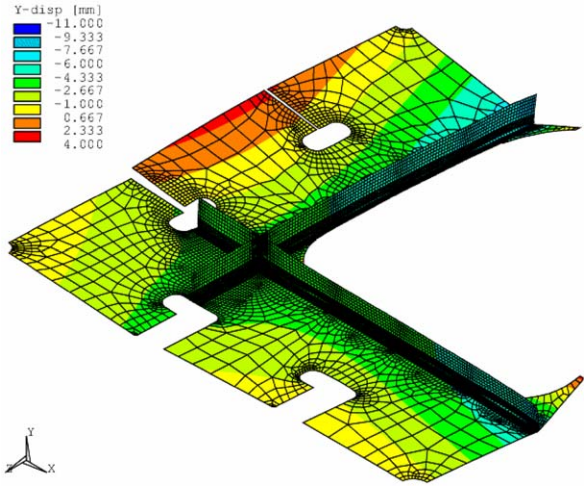


Fig. 6. Deformed plate due to given welding sequence [100].

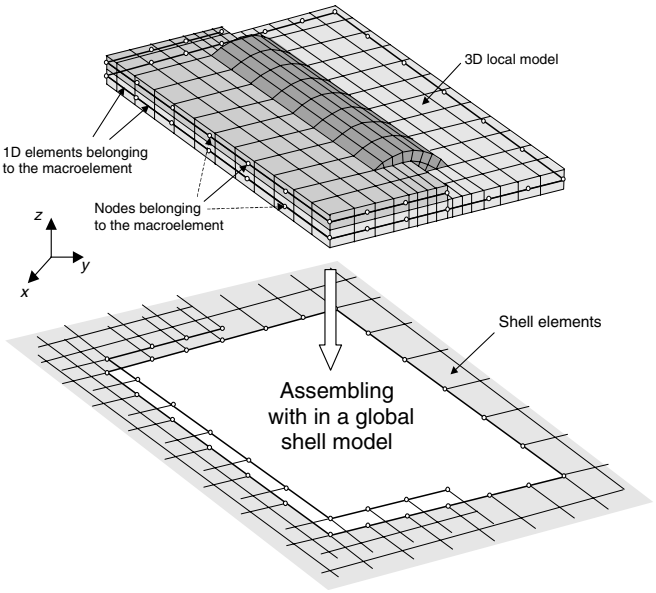


Fig. 7. Macro element definition [101].

Souloumiac et al. [101,102] applied a similar local–global model in combination with an inherent strain approach [103,104]. They assumed that the residual plastic strains of the weld only depend on local mechanical and thermal conditions. However, they underlined the problem of knowing what boundary conditions should be applied to the local model as they influence the computed plastic strains. They also proposed a combination of local, transient macro-elements and steady-state (Eulerian) macro-elements to solve the local problem, Fig. 7. The nodal loads corresponding to the computed plastic strains are then condensed to the boundary of the local model and there they are applied to the global model in order to compute the global behaviour.

6. Modelling aspects of CWM

The scope of the analysis and pertaining accuracy levels are defined and related geometric models are described in the first two sections. Thereafter heat input models, material models and special issues for simulation of multipass welding are discussed. Finally, modelling techniques used to reduce the computational effort are described.

6.1. Scope of analysis and accuracy

The scope of the analysis must be determined before creating a model. ''What question(s) should be answered by the welding simulation?'' This scope of the analysis determines what kind of simplifications that can be done when modelling the welding process. Table 2 lists possible questions in order of increasing complexity. The simulations are accordingly divided into different accuracy categories, Table 3. There is an overlap between the accuracy levels with respect to which questions they are needed for and the higher level should be chosen if details near the weld are needed.

Most welding simulations are used for obtaining residual fields and thus they have models with *basic* or *standard* accuracy. The difference between these levels is that the *standard* accuracy has a finer mesh, smaller time steps and a more detailed material model and heat input model than the *basic* model. The latter denotes models with considerable simpli-

Table 2
Pre-evaluation phase. Questions that may be the scope of a welding simulation

1	What are the residual stresses? These will be used for determination of risk for buckling, fatigue, cold cracking or stress corrosion cracking etc.
2	What are the final deformations? These are wanted in order to check if the tolerance requirements on component are fulfilled after the welding procedure
3	What are the transient stresses and deformations? These may be wanted in order to find procedures that maintain the gap required for a successful weld
4	What is the microstructure of the weld and the zone (HAZ)? This kind of analysis also requires a microstructure model
5	What causes hot cracking? Note that hot cracking phenomena are on the limit of continuum mechanics as it may be caused by liquid material along grain boundaries exposed to tensile strains. The parameters obtained from simulations must be evaluated with care in order to ascertain if the model can be used to study how the welding process can be changed to reduce hot cracking

Table 3
Accuracy levels wanted in a welding simulation

1	*Reduced accuracy levels*. Use of simple and fast models for evaluations at preliminary design stages
2	*Basic simulation* where only the overall residual state is of interest, questions 1 and 2. A standard simulation (next level of accuracy) is required if the structure is more flexible
3	*Standard simulation* where only the residual state is of interest. It is used when questions no. 1–3 are to be answered
4	*Accurate simulation* where the transient strains and stresses are wanted. This accuracy level is needed when questions no. 3–4 in Table 1 should be answered
5	*Very accurate simulation* where the high temperature behavior and the zone near the weld is important. This is necessary in order to answer questions no. 4–5

fications of the high temperature behaviour. This is possible when residual stresses are computed for a ''rigid'' structure, Lindgren [105]. There is no precise definition of what is a rigid structure but a simple guideline is that multipass welds are considered ''rigid'' as the size of each weld is small compared to the dimension of the welded parts. All other structures are ''flexible''. The *accurate* models are similar to the standard models but have a better modelling of the high temperature behaviour in order to obtain the transient history. The *very accurate* model requires very good description of the material behaviour and a fine mesh. Additional models of microstructure and/or criteria for fracture etc may also be needed for these cases [106–108].

6.2. Geometric models

The different types of geometric models and thereby finite elements that are used in CWM are given in Table 4 together with their applicability for different accuracy levels defined in Table 3. See Lindgren [10] for an overview of the use of different types of 2D models and 3D models including some simulations of multipass welding using 3D models. The 2D models have often been used with success for residual stresses and deformations but there are exceptions [109].

6.3. Heat input models

Analytic solutions for the thermal fields have been used in several studies, mostly Rosenthal's solution. These solutions may work well for regions away from the weld. However, they are not appropriate nearer the weld, as the arc cannot be considered as a point source then. Furthermore, the thermal properties vary strongly with temperature. There is no reason to use the analytic solutions any more as the numerical simulation of the thermal field is quite straightforward except for *reduced accuracy* models where quick and simple solutions are needed.

All simulations in CWM consider only thermo-mechanical phenomena in the weld and incorporate all the physics of the welding process into empirical heat input models. Thus the heat input must be calibrated based on experimental observations. See the reviews of Goldak et al. [110–115] and Lindgren [10]. Goldak et al. [111] also includes some discussion of advanced welding process models.

The temperature field can be obtained by specifying the temperature in the weld as the thermal load in the finite element simulation. The weld metal is held for a given time at a high temperature and then the temperature is not prescribed anymore but computed according to the heat conduction equation. Carmet et al. [116] and Goldak et al. [117] used prescribed temperatures. Jones et al. [118,119] prescribed the temperature at the boundary of the weld pool to be equal to the T_{liquidus}. Roelens et al. [120–122] and Lindgren et al. [16] prescribed the temperature in the case of multipass welds. The other possibility to compute the temperature field is to use heat flux as the thermal load. The use of constant prescribed flux with some ramping function was used by Andersson [5] and also many others. Goldak et al. [123] proposed the so-called double ellipsoid heat source that is a more accurate representation of the heat flux. It was later extended to arbitrary distribution

Table 4
Possible accuracy levels for different geometric features of models

Type of geometry	Basic/standard	Accurate	Very accurate
2D-X[a] and 2D-multipass weld[a]	X	X	X
2D-transient[b,c] and 3D-shell[c]	X	X	Cannot resolve a correct stress field near weld
3D-solid	X	X	Still too computer intensive
3D-multipass weld	X	Still too computer intensive	Still too computer intensive

[a] Model of cross-section of weld using plane strain, generalized plane deformation or axisymmetric models.
[b] Models using plane stress elements.
[c] Multipass welds can be accommodated approximately by changing thickness of elements.

functions [113]. Sabapathy et al. [124] modified the model for the case of weaved welding and Lu et al. [125] implemented a version where it could be applied instantaneously over segments. The double ellipsoid heat source has been used in many papers, for example [126–128]. The Gaussian heat source model is appropriate for *accurate* analyses and it needs at least four quadratic elements along one semi axis in order to resolve the Gaussian distribution of heat [115].

6.4. Material models

This section shows how the varying material behaviour over a large temperature range and in the presence of phase changes is modelled in CWM, see also Lindgren [11]. The thermal aspects are quite straight forward whereas the mechanical material behaviour is more complex.

6.4.1. Material modelling in thermal analysis

Some non-standard modelling considerations for the thermal analysis in CWM are related to the weld pool. It is typically assumed that the latent heat is evenly distributed during the solidification/melting. However, Feng [129], Feng et al. [130] and Dike et al. [131] used a more advanced model for the release of latent heat during solidification of aluminium. Their model gives a higher release of latent heat in the upper part of the T_{solidus} to T_{liquidus} interval. The heat in the weld pool is not only conducted but also convected due to the fluid flow. Many analyses imitate this by increasing the thermal conductivity at high temperatures, e.g. Andersson [5], Leung and Pick [126]. Different heat conductivities in different directions can also be used [90,128].

It should also be noted that there are also latent heats to be included when solid state transformations occur.

6.4.2. Material modelling in mechanical analysis

The material has a high temperature during a relatively short time of the weld thermal cycle and therefore the accumulated rate-dependent plasticity is often neglected. This was stated clearly already by Hibbitt and Marcal [2] and others [132–135]. They included rate-dependent plasticity only for the stress relief phase. Berglund [136] found that the viscous effects could be ignored during the welding process and were only important to consider during the holding phase of the subsequent stress relief treatment. Sekhar et al. [137] showed the effect of using yield stress with temperature for two different strain rates. The difference is small for that particular case except around 700–900 °C. However, it has also been found that the high temperature behaviour and thereby also the viscous material behaviour do affect deformations in the case of structures with small restraints [136,138].

Most papers use the common rate-independent plasticity model based on von Mises yield criterion but some papers use rate-dependent plasticity. Argyris et al. [139,140] used an overstress model of the form

$$\dot{\bar{\varepsilon}}^{\mathrm{p}} = \mu \left\langle \left(\frac{\bar{\sigma}}{\sigma_{\mathrm{y}}} \right)^2 - 1 \right\rangle. \tag{49}$$

This is similar to the flow strength equation in Eq. (48) but cannot be rewritten to that form. They used a Ramberg–Osgood relation for the evolution of the yield stress.

Chidiac and Mirza [141] used

$$\dot{\bar{\varepsilon}}^{\mathrm{p}} = AS_0 \left\langle \frac{\bar{\sigma}}{\sigma_{\mathrm{y}}} - 1 \right\rangle^n \mathrm{e}^{-Q/RT}, \tag{50}$$

where A is related to grain size, Q is the activation energy for grain growth and R is the gas constant.

Myhr et al. [142,143] used a viscoplastic material model

$$\dot{\bar{\varepsilon}}^{\mathrm{p}} = \left(\frac{\bar{\sigma}}{A(\text{microstructure}) F(\theta)(\bar{\varepsilon}_0^{\mathrm{p}} + \bar{\varepsilon}^{\mathrm{p}})} \right)^{1/m(\theta)}, \tag{51}$$

which is coupled to models for microstructure evolution affecting hardening. The strain hardening has a starting value $\bar{\varepsilon}_0^{\mathrm{p}}$ and no increase in $\bar{\varepsilon}^{\mathrm{p}}$ is allowed at temperature above a critical temperature.

Wang and Inoue [144] and Inoue and Wang [145] used a Perzyna type of model

$$\dot{\bar{\varepsilon}}^{\mathrm{p}} = \mu \langle \Phi \rangle. \tag{52}$$

Goldak et al. [117,146] discussed the use of different rate dependencies for the plastic behaviour at different temperatures and stresses. They chose to use different constitutive models for different temperature regions. A linear viscous model is used at a homologous temperature above 0.8. Rate-dependent plasticity is used down to a homologous temperature of 0.5 and von Mises plasticity for lower temperatures. Mahin et al. [147,148], Winters and Mahin [149], and Ortega et al.

[150,151] and Dike et al. [152] used a unified creep-plasticity model by Bammann [80,153]. Sheng and Chen [19], Chen and Sheng [17,18] used the Bodner–Partom viscoplastic model and the Walker model. The latter model also accounts for kinematic hardening.

The material near and in the weld is subjected to reversed plastic yielding during the cooling phase. Thus using kinematic, isotropic or combined hardening will affect the stresses in this region. Bammann and Ortega [154] investigated the effect of assuming isotropic and kinematic hardening. They found that the choice of hardening influences the residual stresses very much in the weld metal but further away the different models gave identical results. Devaux et al. [155] found only small differences in residual stresses between models with isotropic or kinematic hardening. They studied weld repair with 4 or 6 beads. Most studies use linear, isotropic hardening. Friedman [3,156] assumed isotropic hardening using a power law hardening. Kinematic hardening (Papazoglou and Masubuchi [157], Wang and Inoue [144], Leung and Pick [126], Tekriwal and Mazumder [158], Michaleris and DeBiccari [128]) and combined hardening (Chakravarti et al. [159,160], Murthy et al. [161]) have been assumed in some simulations.

Further details about models and data are given in the review [11]. However, the problem is not a shortage of models but the lack of high temperature data. This is a problem even for the simplest plasticity models and even more for more complex models. This fact in combination with the approximations of the weld pool behaviour discussed earlier and the numerical problems that may be caused by a too low stiffness in the high temperature region have led to the notion of a *cutoff* temperature in the field of CWM. No changes in mechanical properties are accounted for above this temperature. Several investigations have been done regarding the effect of the choice of cutoff temperature, see [11]. Tekriwal and Mazumder [158] varied the cut-off temperature from 600 °C and up to the melting temperature. The residual transverse stress was overestimated by 2–15% when the cut-off temperature was lowered. A cutoff can be used up to *accurate models* in Table 3. However, care should be taken when deformations for a loosely fixtured weld. Then the high temperature behaviour may be important [136,138].

The material behaviour does not only depend on the current temperature but also on the current microstructure. The thermal driving forces, coupling 5 in Table 3, are usually assumed to cause the microstructure evolution and coupling 6 is ignored. Then the material behaviour depends on the current temperature and its history. There are three possible approaches to handle this

- Ignore microstructure changes.
- Account for them in a simplified manner.
- Perform a TMM analysis.

The first approach is used for copper [89], stainless steels [162] or Inconel [3] where phase changes are ignored. Phase changes have also been ignored in the case of ferritic steels in some papers [1,2,16]. However, even the *basic accuracy* models must at least account for the influence of the phase changes on the final yield limit of the material [105] but other effects of phase changes may be ignored for these models. The second approach usually only includes the influence of the phase changes on thermal dilatation and yield limit. Andersson [5] and others [51,90] used the peak temperature and the cooling rate from 800 to 500 °C as variables determining the microstructure of the material and thereby what material properties should be used during the cooling. The TMM-approach, requires a model for the microstructure evolution and a model for mixing the material properties of each phase. Börjesson and Lindgren [30] and Roelens [122] investigated multipass welds using a TMM models. Grøng [9] described microstructure models for aluminium where precipitates are important for the strength.

The additional material properties to be considered in the presence of phase changes are

- volume changes due to phase changes,
- annealing and
- transformation plasticity.

The volume changes due to phase changes are included in the thermal dilatation together with the thermal expansion [11]. They are usually ignored for the solid–liquid phase changes, except for very *accurate models*, but often included for solid-state phase changes.

The annealing effect is here denoting the change in dislocation structure due to phase changes, which reduces the hardening. This is modelled as a reduction or removal of accumulated plastic strains. Many analyses remove these completely at melting [89,148] but some also do this during the solid-state transformations [3,156,157]. Devaux et al. [155] argued that it is reasonable that the memory of previous plastic deformation disappears for all solid state phase transformations in ferritic steels with maybe the exception for the martensite formation. The latter involves very small displacements of the atoms during the transformations, which they assumed did not affect the dislocations.

Transformation plasticity is the macroscopic effect of plastic yielding in a weaker phase when a phase with higher yield stress is formed. Typically, transformation plasticity occurs when austenite forms to bainite or martensite. Many researchers in CWM have modelled transformation plasticity as a deviatoric plastic component that is a function of applied stress and rate of phase changes. The most commonly used relation is based on the papers by Leblond et al. [163,164] and has the form

$$\dot{\bar{\varepsilon}}^{\mathrm{tp}} = K\frac{\mathrm{d}\varphi}{\mathrm{d}X_{\mathrm{M}}}\dot{X}_M\bar{\sigma}, \tag{53}$$

$$K = \frac{5}{6}\frac{\sqrt{\left(\varepsilon^{\mathrm{tr}}_{\gamma\to\mathrm{M}}\right)^2 + 3/4\left(\gamma^{\mathrm{tr}}_{\gamma\to\mathrm{M}}\right)^2}}{\sigma_{\mathrm{y}^*}}, \tag{54}$$

$$\sigma_{\mathrm{y}^*} = \sigma^{\mathrm{M}}_{\mathrm{y}}\left(\frac{1-\sigma^{\gamma}_{\mathrm{y}}/\sigma^{\mathrm{M}}_{\mathrm{y}}}{\ln\left(\sigma^{\gamma}_{\mathrm{y}}/\sigma^{\mathrm{M}}_{\mathrm{y}}\right)}\right), \tag{55}$$

where $\varepsilon^{\mathrm{tr}}_{\gamma\to\mathrm{M}}$ is the transformation volumetric strain, $\gamma^{\mathrm{tr}}_{\gamma\to\mathrm{M}}$ is transformation shear strain, $\sigma^{\gamma}_{\mathrm{y}}$ and $\sigma^{\mathrm{M}}_{\mathrm{y}}$ are the yield limits of austenite and martensite phases, and φ is a heuristic function. This additional plasticity term is only included during elastic processes. This is the case when the stress state is inside the macroscopic yield.

6.5. Modelling of multipass welding

The modelling of multipass welding poses some additional problems compared with single-pass welds. The material modelling can be quite difficult, as discussed in Section 6.4, in the case of repeated phase changes. The addition of filler material must be handled by some kind of a deactivation and activation procedures, Lindgren and Hedblom [94].

It is possible to lump weld passes and ignore some of the thermal history for *basic accuracy* models as found by many investigators, see the discussion about temporal reduction in Section 6.6. Higher accuracy models need each weld to be modelled separately in order to obtain correct transient results. *Accurate* and *very accurate models* need many elements over the cross-section of the weld whereas it suffices with one element for the *standard accuracy* model.

6.6. Modelling for reduction of size

The use of different computational techniques for enabling the solution of large problems was discussed in earlier. Different modelling options to simplify the welding simulations are outlined below.

Dimensional reduction is the most common way to reduce the computer time. Simplifying the geometry and neglecting heat conduction in welding direction are the basic assumptions for 2D-X models. This corresponds to assuming infinite welding speed. The case studied by Dike et al. [109] is an interesting example where the assumption of infinite welding speed was inappropriate leading to a need for a 3D model. There have been published a large number of papers using 2D-X models and also plane stress models, see Lindgren [10].

Berglund and Runnemalm [165] compared the use of 2D-X, 3D-shell and 3D-solid models of a butt-welded plate. They found, as also McDill et al. [166] found, that the residual stresses in the longitudinal are higher for the plane strain model. The 2D-X model cannot represent all deformation modes but can give quite good residual stress fields. The shell elements can, pertaining the thermal load is applied correctly, give overall good behaviour. Many simulations have also used shell elements in the weld regions despite the fact that the shell assumptions are not valid there [33,167–170]. Even multipass welds and repair welds have been simulated by changing the shell thickness. Another possibility is to apply shell and beam elements at appropriate places in the regions further away from the weld and 3D elements in the weld. Gu and Goldak combined them in a thermal analysis of a weld [171] and Näsström et al. [172] performed a thermo-mechanical analysis of a weld. McDill et al. [173] developed an graded shell-solid element for to be used in welding, This approach paves the way for combining adaptive meshing and element switching with shell and solid elements. Thus the stress and deformation pattern together with the element geometry can be used to adaptively determine whether an element should be a solid or a shell.

Another strategy for dimensional reduction is to imitate the real 3D behaviour of a structure by adding discrete stiffnesses. This failed in the case studied by Dike et al. [109]. Rybicki and Stonseifer [174] added stiffnesses orthogonal to a 2D model of welding in order to include 3D effects. These stiffnesses were obtained from a 3D model of the structure. Michaleris et al. [175] compared 2D and 3D models of a multipass butt-welded plate. They simulated the first pass with both models. The 2D model had a larger fusion zone, despite a reduced heat input, and a larger zone with high tensile longitudinal stresses than the 3D model. They included an additional dashpot in the 2D model to make it match the 3D results

for the first weld pass. This gave quite good agreement between measured and computed displacement of the butt joint's edge until the plate was turned and the welding continued on the other side.

Temporal reduction is another aspect of the reduction in computer time by permitting longer time steps or shrinking the history of the process. It is usually combined with dimensional reduction and simplification of material modelling. The most common one is the assumption of infinite welding speed leading to a 2D model as discussed above. However, there are also other types of temporal reduction techniques. Early 2D models of multipass welding ignored completely the transient history and used so-called temperature envelopes when computing the residual stress state [176] and/or lumped welds together [177]. Some simplifications of the welding history can lead to symmetric models [178]. The moving, transient heat source can be replaced by a faster source or by a prolonged source [179] where the heat input is extended over a longer region in the welding direction.

Material modelling simplifications are usually combined with the above simplifications, as there is no use of a detailed material model of the high temperature region when this is considerably simplified in the modelling.

Weld substitution is the replacement of the weld model by a corresponding load. Michaleris and DeBiccari [128] evaluated the effect of welding on a 3D model by using a 2D model for the welding simulation and transferring the welding residual stresses to a 3D model of the structure in order to study weld-induced buckling. They created the corresponding load on the 3D model by using anisotropic heat expansion to create an approximation of the longitudinal and transverse stresses obtained in the 2D analysis. The inelastic strains due to the welding are called inherent strain by Ueda and coworkers [103,180]. They assume that these plastic strains only depend on local conditions and can therefore be computed for fewer cases and then be used in different applications as some kind of elementary case, see [104,180,181]. Yuan and Ueda [181] investigated the inherent strain for welded T- and I-joints. The inherent strains were then used as equivalent loads in an elastic analysis as a method for predicting the residual stresses. This superposition of the loads corresponding to the inherent strains can also be combined to multipass welds. The problem of determining the relation between the inherent strain and the welding parameters is addressed by Wang et al. [182]. The effect of restraint due to overall stiffness of the studied geometry on the local behaviour of the weld can be considerably. Murakawa and coworkers [183–185] used the inherent strain method and a gap element for studying the effect of welding sequences on global deformation and end cracking. The gap element is an elastic element used instead of nonlinear contact conditions between the plates to be joined.

7. Applications

The early welding simulations were mostly concerned with predicting residual deformations and stresses. However, this knowledge was seldom used to analyse the effect on the component. Radaj [186] discussed the use of results from welding simulations for different purposes and found it lacking in many respects. We will describe some recent papers where this has been done and also other contributions towards the industrial use of welding simulations. Thereafter contributions to advancing the understanding about how to model welding are described. See the review by Lindgren [10] for publications before 2000. This review only includes later papers that include both thermal and mechanical parts of the welding simulations.

7.1. Using welding simulations

Narrow gap welding procedures with reduced tensile residual stresses and thereby risk for intergranular stress corrosion cracking (IGSCC or SCC) were studied by Fricke et al. [187] and Hedblom [188–191]. U-seam and a narrow gap welding procedures for two different pipes were compared. The latter method gives reduced residual axial stresses on inside of the smaller pipe, Fig. 8, but the improvement was reduced for the larger pipe. Mochzuki et al. [192] studied a circumferential weld in a pipe and changed the order of weld sequences in order to reduce the risk for stress corrosion cracking. They compared two different welding sequences for a multi-pass fillet weld with respect to fatigue life. They computed and measured residual stresses. However, they did not use any model to evaluate their effect on fatigue life. This information was instead obtained from experiments. They found that there was no effect of welding residual stresses when loaded with a high cyclic stress but a clear effect when loaded with low cyclic stresses. The first was probably due to relaxation of the welding residual stresses during the first load cycles. They also studied crack propagation in the presence of welding residual stresses as well as method to reduce the residual stresses. Zhang et al. [193] studied crack propagation in the presence of welding residual stresses due to a circumferential weld in a core shroud. The initial cracks were assumed to be caused by IGSCC and computed stress intensity factors were evaluated to determine how they could grow. A similar phenomenon due to helium bubbles at grain boundaries was studied by Hashimoto et al. [108]. The helium bubbles were created by the radiation of the stainless steel.

Sabapathy et al. [124] investigated the burn-through when welding a pipe with internal pressure. They calculated the radial deflection of the weld pool region during welding versus increasing internal pressure and estimated the limiting pressure this way. They reduced the size of the model by considering a segment of the pipe as it is only the region around the

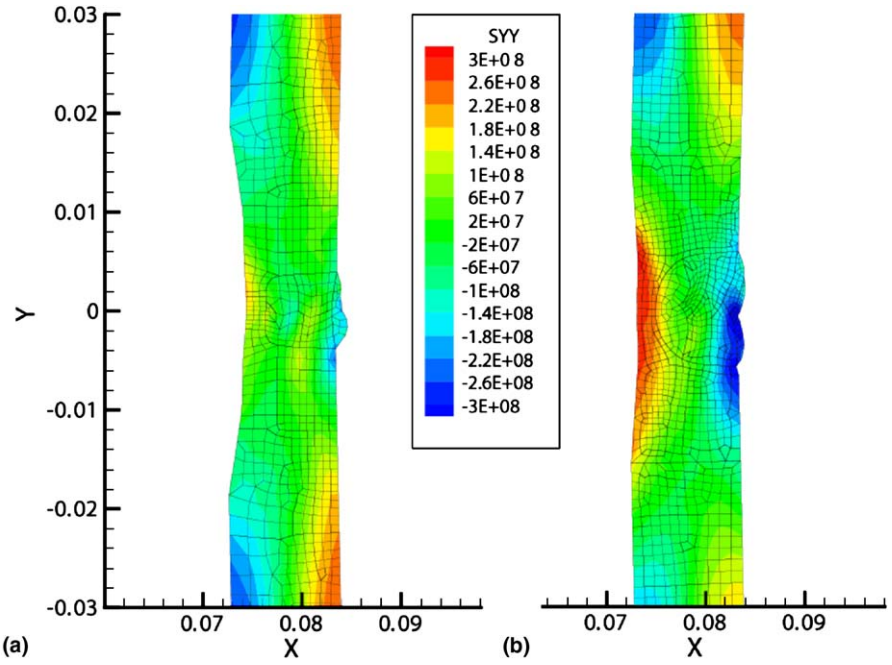


Fig. 8. Residual axial stresses (SYY) due to circumferential weld of two pipes. A narrow gape weld (a) and a U-seam weld (b). A cross-section of the wall around the weld region is shown. The inside of the wall is to the left.

weld that is critical. Teng et al. [194–196] performed finite element analysis of multipass welds in order to obtain residual stresses. They used this for fatigue analysis of critical areas. Hansen [197,198] simulated the welding of a frame box and performed a linear elastic fracture mechanics (LEFM) analysis where the welding residual stresses were included as a fixed mean stress. The residual stresses were found to be beneficial for this case. They concluded that previously applied stress relief treatment made the life time of the component shorter. This was confirmed by experiments. Michaleris and coworkers [199–202] developed welding procedures with minimised distortions. They also developed an efficient optimisation strategies combining sensitivity analysis with an Eulerian approach [38,39]. The design sensitivities required for the optimisation were developed analytically using direct differentiation. A demonstration of the approach was done for plate forming using a laser. The angular of a plate should be maximised subjected to a constraint on the maximum temperature to avoid phase changes. The design parameters were the speed, radius and heat input of the laser. The longitudinal stress for the found optimum design is shown in Fig. 9. Low stress low distortion (LSLD) [203] welding techniques have been designed using welding simulations [204,205]. Simulations have also been used to reduce risk for cracking [206].

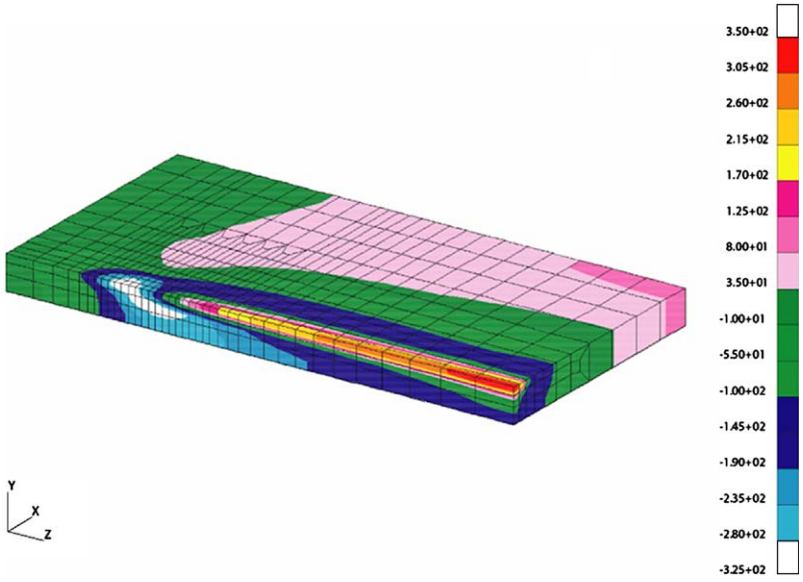


Fig. 9. Longitudinal stress at optimum design [39].

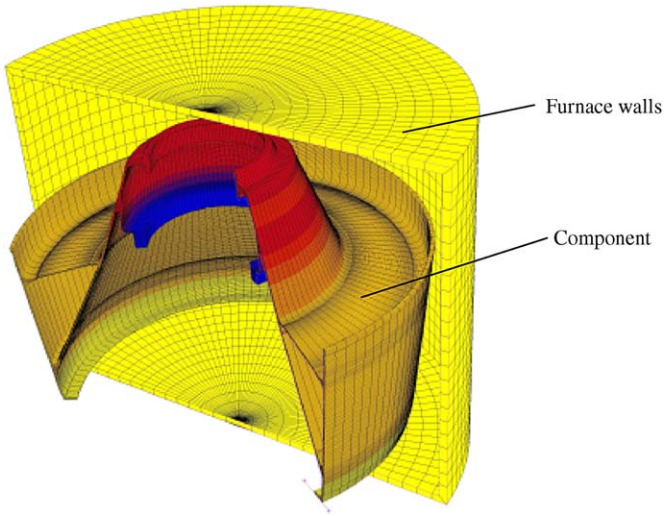


Fig. 10. Thermomechanical model including furnace wall to compute viewfactors for radiation during heating phase.

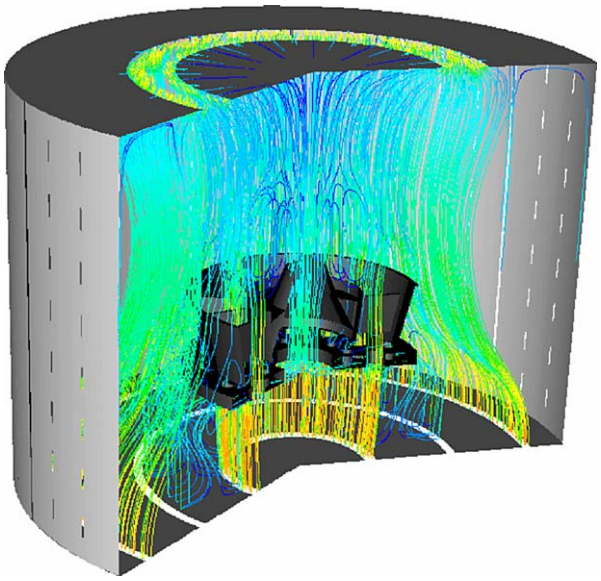


Fig. 11. Stream lines during flow of cooling gas of welded aero-engine component [135].

Hansen [197,207] combined flame cutting and welding simulations for a test plate and compared with measured residual stresses. The plate was of the size 480×480 mm^2 after welding and the thickness was 10 mm. The effect of the flame cutting was not only noticeable near the edges but also in the interior of the plate.

Berglund et al. [135] performed combined welding and heat treatment simulations. Radiation from furnace walls during heating phase was computed by including the wall in the thermomechanical model, Fig. 10. Convective heat losses during the cooling phase in the heat treatment furnace were obtained from a separate CFD analysis, Fig. 11. This was applied on an aero-engine component in order to predict its distortion due to welding and subsequent stress relief treatment. The residual effective stress is shown in Fig. 12.

7.2. Improvement of modelling in CWM

The problem of modelling in CWM is mostly concerned with heat input and material model aspects. Some papers that have contributed to an increased understanding about how to simulate welding are given below.

Stone et al. [208] studied EB-welding of a plate and compared the residual deformations with measurements. They varied the focus of the beam creating varying cross-sections of the weld metal and performed a number of tests. They measured the curvature of the plate along and transverse to the weld and compared with simulations. They did not even obtain

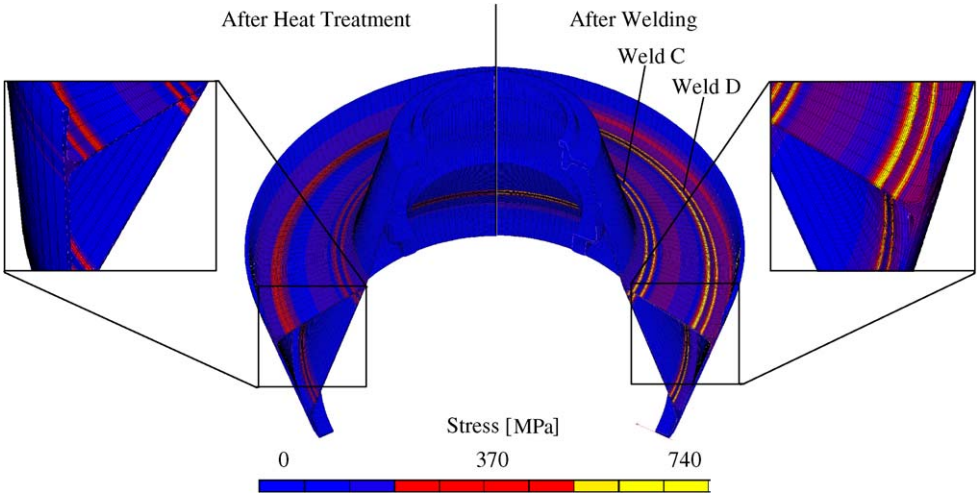


Fig. 12. Von Mises effective stress after welding and heat treatment [135].

the correct sign of the curvature along the weld. Buckling was probably a reason for the discrepancy between measurements and simulations in Stone et al. [208] making this sensitive to small perturbations in the geometry. This explanation was not proven in their study but the similarity with the study by Berglund [136] makes this probable. Berglund found that if the initial geometry (before welding) was included in the model, then the simulations could predict this kind of deformation behaviour.

Lundbäck [24] used a detailed model of the physics of the EB-welding process to calibrate the heat input model. Lu et al. [209] combined fluid flow models in the weld pool with a solid mechanics model using ANSYS. This was done earlier by Sheng and Chen [18,19] but few have developed models based on this combined approach.

Hemmer et al. [92] studied EB-welding of duplex stainless steels. They combined microstructure models for austenite dissolution, subsequent grain growth in delta-ferrite and its decomposition to austenite during cooling. The latter was assumed to be controlled by nitrogen diffusion in the delta-ferrite. This was used to predict austenite depletion in the heat affected zone which can reduce corrosion resistance and toughness. They used a simple model to predict weld penetration and they had a cylindrical heat source in the numerical model that was calibrated by comparison with measured temperatures. Myhr et al. [210] developed a model for computing precipitate distributions [211] for simulation of welding and heat treatment of Al–Si–Mg alloys. Berglund et al. [135] used a simple microstructure model for austenite–martensite transformation of a martensitic stainless steel in their welding simulations.

8. Discussions

8.1. Increased industrial use

The developments since the review in 2000 [10–12] have been geared towards industrial applications in nuclear industry and aerospace but also in other heavy industries [212,213]. The computer hardware and software development is reducing the time needed for welding simulations [214] and it will probably be possible to perform 3D simulation of multipass welding using sufficient fine mesh needed for *accurate* or *very accurate* analyses [105] during the next five years. Commercial software has often been used for welding simulations. This has required the implementation of user routines in order to enable welding simulations. However, the code developers are implementing more support for CWM into their codes. This commercial interest is due to the increased industrial use of welding simulations. The author has in his industrial cooperation seen the use of simulation for avoiding cracking, controlling deformations and stresses by means of optimal welding procedures and even seen how simulations have been used in business to promote high tech products.

This industrial use has created a need for a formulated methodology for welding simulations [105] and validation procedures [136,215] to enable engineers to perform welding simulations. The calibration of heat input and material models is closely related to validation procedures. Berglund [136] proposed a procedure based on the concepts in Oberkampf [14]. The validation has to be done for the heat input and the material models first. Some experimental observations are used to calibrate the heat input model and others for its validation. These can be different sets obtained from the same welding experiment. Tensile tests for calibration and the Satoh test [216] for validation of the material model has also been a common strategy [215,217]. Finally, the complete validation of the model can be done using a sub-system model where thermal

and mechanical response is validated for a given finite element model. Thereafter the model is used to predict the response of whole systems.

Welding is always performed in combination with other manufacturing processes in industry. Berglund et al. [135] studied welding and stress relief heat treatment and it is to be expected that more work involving several process steps will be published in the future. This will also make it possible to optimise the overall manufacturing chain in addition to individual processes as Michaleris et al. [200,202,218] have done.

8.2. Improved modelling

There is a need to improve or even remove the need for heat input models in order to connect the heat input into the finite element model with the actual welding parameters for different processes. One possibility could be to combine thermo-mechanical models with fluid flow models for the weld pool of the type used by Amberg and co-workers [219,220] or DebRoy and co-workers [221–224]. Furthermore, models are needed for the specific heating process as in Zhu [23] for a flame or by Sudnik et al. [22] for a laser.

The modelling requirements for *basic* or *standard* analyses are quite established. However, there will be a continuing need to improve material modelling and its coupling to the microstructure evolution for more detailed analysis of the weld behaviour and pertaining questions about, for example, cracking. The use of more physical based model [225] for material behaviour and its relation to the microstructure as well as evolution equations for the microstructure [226] are expected to improve the predictive capability in CWM.

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